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Paleomagnetic and geological investigation into southern Sinai volcanic rocks and the rifting of the Gulf of Suez

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Abstract

Two volcanic occurrences from southern Sinai were sampled for paleomagnetic, petrographic and geochemical investigations. 29 oriented samples were taken from six sites of the Middle Jurassic basaltic intrusion at Wadi Budra, southwest Sinai (28°55′N, 33°20′E). After alternating field (AF) demagnetization and tilt correction, all samples exhibit normal magnetization with a mean paleomagnetic direction $D=63.3°$, $I=8.0°$ and $\alpha_{95}=10.2°$. This yields a paleopole position at lat. 25°N, long. 133°E and $A_{95}=8.6°$. Also, six oriented blocks (34 core samples) were taken from the Sant Catherine doleritic dike at Wadi Sahl el Raha, south Sinai (28°33′N, 33°55′E). After AF demagnetization all sites showed normal polarity with a mean direction of magnetization $D=3°$, $I=39°$ and $\alpha_{95}=15.3°$ and a paleopole position at lat. 83°E, long. 179°E and $A_{95}=16.5°$. The paleomagnetic pole of W. Budra is found to be incompatible with poles of similar age from Africa. The cause of this cannot be uniquely determined, but most probably could be referred to local tectonic rotations and/or geomagnetic secular variations affecting the paleomagnetic directions of these samples. In contrast, the Tertiary pole of Wadi Sahl el Raha is consistent with the part of the apparent polar wander path (APWP) of Africa for the time period 20 Ma (Oligo-Miocene age).

Petrographical and geochemical investigations revealed that the volcanic rocks of both localities are continental within-plate, rift-related alkali olivine basalt (W. Budra) and olivine dolerite (Sahl el Raha). The primary magnetite is the main opaque mineral present in the samples, which represents the main carrier of magnetization in these rocks.
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1. Introduction

In the classical models the relation between the uplift, rifting and volcanism explained the Red Sea rift as a result of tension on the crest of the up-doming Arabian–Nubian Shield. However, in modern concepts of plate tectonics, the present tectonic setting of the Red Sea has been used as a model for the Proto-Atlantic — an early stage of crustal separation. This concept implies horizontal separation and relative rotation of Africa and Arabia. The amount of separation, the mechanism of movement and the number of crustal plates involved vary widely in the various models proposed. They fall mainly into two categories: (a) those assuming crustal separation of the Red Sea over its entire width, joining remarkably well-

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fitting shorelines (Swartz and Arden, 1960; McKenzie et al., 1970); and (b) those assuming crustal separation over part of the present width of the Red Sea only (Phillips, 1970; Le Pichon et al., 1974).

In addition, Drake and Girdler (1964) proposed a model that assumes crustal separation in the central graben (axial trough) only. In general, the rifting of the Red Sea and associated system was initiated in the Oligocene, with thermally driven uplift and domal arching of the Arabian–African Shield crust (Bannert, 1971). Two triple junctions characterize the Red Sea system, the Afar to the south and the Sinai to the north. As a result of rifting, the Arabian plate separated and moved NE to collide with the Eurasian plate along the Taurus–Zagros mountain belt.

During the initial phase of rifting, numerous basaltic dikes intruded the whole of the Red Sea and the Gulf of Suez with orientations parallel (and subordinately also oblique) to the long axis of the rift. The intrusion most probably was emplaced during a single event at about 20 Ma (Early Miocene); all dikes, regardless of their distance to the central rift zone, are of almost the same age, so that a diffuse intrusion mechanism may be assumed (Baldridge et al., 1991). On Sinai the dikes can be found in the Precambrian basement as well as in the Phanerozoic sediments. It is assumed that the basaltic magmas originate from the lithospheric mantle (Baldridge et al., 1991; Eyal et al., 1981).

The important geodynamic process is the counterclockwise rotation of the Arabian plate relative to Africa, which led to the Late Oligocene–Early Miocene opening of the northern Red Sea, including its northern extension, the Gulf of Suez. The Gulf of Suez soon became a failed rift, probably related to the process of collision of the Arabian and Turkish plates, which also initiated the opening of the Gulf of Aqaba.

The geographic position of the Sinai subplate holds the key to a proper understanding of the tectonic interaction between the Arabian plate, the African plate and the Levantine subplate. Paleomagnetic investigation of the volcanic rocks of Sinai is a potentially fruitful approach because of its relevance to the paleotectonic development of Sinai itself, as well as the tectonic development of the Red Sea and the Gulf of Suez rift system. Until now, few paleomagnetic data have been published from Sinai (Wassif, 1988, 1989, 1991; Abdeldayem et al., 1994; Abdeldayem and Tarling, 1996; Kafafy et al., 1996; Ibrahim et al., 1998). Therefore, the extension of these paleomagnetic studies to cover all the Phanerozoic volcanic rocks from Sinai is very important in proposing a dynamic model for the origin of these volcanic rocks and intrusions and their tectonic relation to the opening of the Red Sea and the Gulf of Suez rift.

In the present study, petrographic, geochemical and paleomagnetic investigations are carried out on two volcanic occurrences from southern Sinai (Fig. 1). The first occurrence is an olivine basaltic intrusion at Wadi Budra, southeast Abu Zeniema, which was dated as Middle Jurassic (182 ± 7 Ma; Meneisy, 1986; Saleeb Raufail et al., 1989). The second is in Sahl el Raha area from the Sant Catherine NW-trending dolerite dike which extends from Wadi Kid in the Gulf of Aqaba to Abu Zeniema in the Gulf of Suez. The age of the doleritic dike is Oligo-Miocene (21 Ma; Steinitz et al., 1981), which is related to the opening of the Red Sea rift.

2. Geological setting

The Phanerozoic volcanicity and igneous activity in Egypt occurred mainly along the structural features and fracture systems that originated in the late Precambrian (El-Shazly, 1977). De Gruyter and Vogel (1981) and Meneisy (1986) suggested that the alkaline complexes and other plutonic and volcanic assemblages related to this Phanerozoic activity were emplaced along reactivated Pan African fractures or pre-existing zones of weakness.

In Sinai, the Phanerozoic volcanics are represented as sporadic; but frequent, lava flow, sills, dikes and small irregular stocks (Mousa, 1987). Some occurrences of these volcanics are located at Gebel Farsh el Azraq, Wadi Abu Natash and Wadi Budra and its continuation at Nagb Budra. Similar volcanic occurrences are reported in the Western

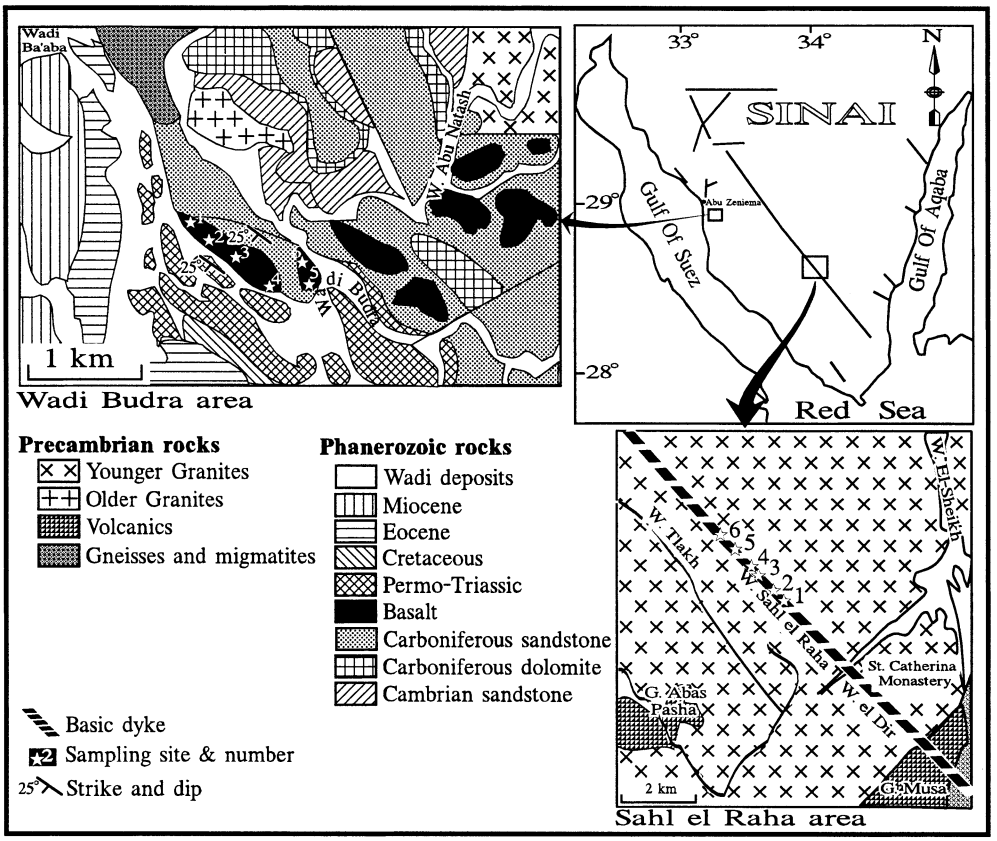


Fig. 1. Geologic maps of Budra area (after El Agami, 1996) and Sahl el Raha area showing the location of the sampling sites.

Desert of Egypt, namely the basaltic dike at Owienate area (southwest Western Desert) of Early Triassic (Meneisy, 1986) and in the Camel pass area of the northern Western Desert of Late Triassic (Abu El Ata, 1988). These volcanic phases were roughly coincident with the Hercinian (Variscian) orogeny in Europe, and the creation of the supercontinent of Pangaea from the northern and southern supercontinents of Laurasia and Gondwana, respectively (Morgan, 1990). These volcanic phases predate any tectonic activity connected with the rifting of the Red Sea (Meneisy, 1986).

Another magmatic phase resulted mainly in a system of long NW–SE-trending basaltic dikes, parallel to the Red Sea and the Gulf of Suez (e.g. Abu Durba, Sharm el-Sheikh, Sant Catherine, Gental). These volcanic rocks are dominantly basaltic in composition and were emplaced during a regional magmatic event related to the initial stage of the formation of the Red Sea and the Gulf of Suez (Eyal et al., 1981; Coleman and McGuire, 1988; Saleeb Raufail, 1989). According to Saleeb Raufail (1989), the relationship between these basaltic rocks and the Gulf of Suez rift is evident. Garfunkel and Bartov (1977) added that the rifting of the Gulf of Suez was well underway in the Late Oligocene to Early Miocene. Generally, the Phanerozoic volcanic rocks are diversified in their chemical characteristics, size and mode of eruption during the Paleozoic and Mesozoic time, whereas during the Tertiary and Quaternary they are largely basaltic (Meneisy, 1990).

In this study, the samples were taken from two volcanic occurrences in southern Sinai. Their geological setting and stratigraphic location were discussed by many authors. The first occurrence is represented by a basaltic intrusion of 5–20 m thick

(Mousa, 1987) at Wadi Budra (Fig. 1). This basaltic intrusion separates the underlying Ataqa Formation (Kostandi, 1959), which was recently named Abu Thora Formation (Kora, 1995), of Carboniferous age, from the overlying Budra Formation, which was named Quiseb Formation, of Permo-Triassic (Issawi and Jux, 1982) or Triassic age (Druckman et al., 1970; Beyth, 1981). At the sampling locality, it is noticed that Abu Thora Formation is metamorphosed at its contact with the basaltic rocks, and its upper part is characterized by a hard quartzite layer. In contrast, the contact between this volcanic rock and the overlying Quiseb Formation is very difficult to identify in the field. These volcanics were K–Ar dated to 182 ± 7 Ma, corresponding to Middle Jurassic (Mousa, 1987). This dating and the presence of a similar basaltic intrusion between the Lower Carboniferous (Abu Thora Formation) and the Upper Carboniferous (Abu Durba Formation, El Agami, 1996) at Wadi Um Themium, 7.5 km SE of the sampling locality, indicate that this basaltic rock is a sill rather than a flow. Based on the field criteria, the basaltic occurrence at Wadi Budra may represent a Late Paleozoic or Early Mesozoic volcanic event. Structurally, the basaltic sill, overlying Abu Thora and underlying Quseib Formations, is inclined with attitude 130°/25°SW.

The second volcanic occurrence is at Sahl el Raha area, which is part of the Sant Catherine doleritic dike that crosses the Sinai Peninsula from Wadi Kid near the Gulf of Aqaba coast and directed NW to Abu Zeniema, passing through Wadi el Dier, Sahl el Raha and Wadi Sahab (up to 100 km in length and 100 m in width). This dike intrudes the country rock of the Younger Granite at the sampling locality (Fig. 1) and is characterized by its negative topography and spheroidal structure.

3. Petrography

The volcanic rocks from the two sampled localities show to some extent similar petrographic characteristics, except that Sahl el Raha dolerite is medium-grained whereas Budra basalt is fine-to medium-grained. The rocks are composed essentially of plagioclase, mafic clinopyroxene and olivine, and Fe-oxide accessory minerals (Table 1) as confirmed by X-ray diffraction (XRD) analysis (Fig. 2). Texturally, the samples possess typically intergranular, subophitic, ophitic and rarely amygdaloidal textures.

In both volcanic occurrences, plagioclase is non-oriented lath-shaped crystals of andesine to labradorite composition (An_{44-65}), with different sizes in relation to its locality. Plagioclase is partially to completely embedded within pyroxene, forming ophitic and subophitic texture, and occasionally interlocked to form intergranular texture. Olivine is commonly altered and usually pseudomorphs to a mixture of reddish brown iddingsite and/or green bowlingite. The occurrence of these two varieties of olivine alteration depends upon the state of oxidation (Deer et al., 1992; Shelley, 1993). In contrast, diopsitic clinopyroxene is almost unaltered and forms coarse poikilitic crystals up to 3 mm, with c^Z angle of 40–45°, with abundant plagioclase and iron oxide inclusions. Opaques are mainly subhedral to anhedral with a very minor euhedral and square crystal habit. These opaques are enclosed within both plagioclase and pyroxene. Furthermore, in the case of Budra basalt, uncommon sphurolitic zeolites are present as amygdals fill. The mineral paragenesis of the two volcanics indicates that the iron oxides and olivine were crystallized first, followed by plagioclase and finally diopside.

The modal mineralogy of these rocks was confirmed using the XRD technique (Fig. 2) in order to confirm the primary mineralogy and to determine the Fe-oxide minerals responsible for carrying the magnetization in these samples. In order to determine the Fe-oxide minerals, the samples were subjected to systematic mineral separation using heavy liquids and magnetic fractionation. The opaque fraction and/or hand-picked minerals were found to be magnetite as main magnetic mineral, with a minor amount of ilmenite. Hematite is present as an oxidation product of magnetite and ilmenite.

4. Geochemistry

Six representative samples (three samples from each locality) were analyzed for major oxides using

Table 1
Major oxides (wt%), trace elements (ppm), normative and modal minerals of the studied volcanic rocks

Locality	Budra Middle Jurassic volcanics				Sahl el Raha Oligo-Miocene dike			
S. No.	3	4	6	Mean	2	3	5	Mean
(a) Major oxides (wt%)								
SiO_2	46.82	47.20	45.37	46.46	48.40	47.27	47.10	47.59
TiO_2	1.98	1.35	1.49	1.61	2.05	2.23	3.22	2.50
Al_2O_3	16.96	16.84	17.07	16.96	15.52	15.07	15.19	15.26
Fe_2O_3	4.31	3.34	5.04	4.23	5.46	5.43	5.50	5.46
FeO	6.34	6.96	7.04	6.78	6.03	6.85	6.35	6.41
MgO	7.99	8.70	8.80	8.50	6.45	6.85	7.03	6.78
CaO	9.22	8.64	8.56	8.81	8.46	9.07	8.54	8.69
Na_2O	4.05	3.93	4.04	4.00	3.85	3.84	3.92	3.87
K_2O	0.46	0.45	0.89	0.60	0.98	0.73	0.93	0.88
P_2O_5	0.30	0.58	0.50	0.46	0.73	0.58	0.43	0.58
H_2O	1.58	1.97	1.19	1.58	2.04	2.10	1.76	1.97
Total	100.01	99.96	99.98	99.98	99.97	100.02	99.97	100.02
(b) Trace elements (ppm)								
Cu	40	44	43	42	39	49	44	44
Ni	96	85	51	73	71	76	69	72
Co	125	122	115	121	123	122	128	124
Cr	148	146	132	142	142	144	140	142
V	184	179	101	155	259	293	269	274
Zn	124	130	124	126	122	124	116	121
Ga	18	16	23	19	17	18	19	18
Pb	154	129	249	177	118	165	183	155
Ba	293	253	181	242	360	457	432	416
Zr	219	214	226	220	211	220	220	217
Y	47	45	51	48	43	47	47	46
Sr	276	279	270	275	287	262	260	270
Rb	87	81	101	90	77	88	91	85
Nb	34	33	35	34	31	34	33	33
(c) Normative minerals								
K-feldspar	–	–	–	–	5.90	4.40	5.62	5.31
Albite	30.23	32.46	26.14	29.61	33.25	33.20	33.77	33.41
Anorthite	27.15	27.51	26.18	26.95	22.65	22.17	21.48	22.10
Leucite	2.18	2.18	4.36	2.91	–	–	–	–
Nepheline	2.49	0.80	4.53	2.60	–	–	–	–
Diopsite	13.83	9.93	10.78	11.51	12.16	15.89	14.81	14.29
Hyperthene	–	–	–	–	8.25	1.07	–	3.10
Olivine	13.23	18.21	16.88	16.11	4.02	9.49	9.01	7.51
Magnetite	6.34	4.93	7.39	6.22	8.08	8.03	8.13	8.08
Ilmenite	3.82	2.61	2.87	3.10	3.98	4.32	6.22	4.84
Apatite	0.69	1.37	1.18	1.08	1.73	1.37	1.01	1.37
(d) Modal minerals								
Plagioclase	49.5	47.5			51.3		58.6	
C. pyroxene	33.3	26.1			26.6		27.7	
Olivine (total)	12.7	21.6			15.5		7.4	
Fresh	0.9	5.5			1.8		2.2	
Iddingsite	3.0	13.1			7.7		4.9	
Bowlingite	8.8	4.4			9.2		0.6	
Iron oxide	4.3	3.6			3.4		6.0	

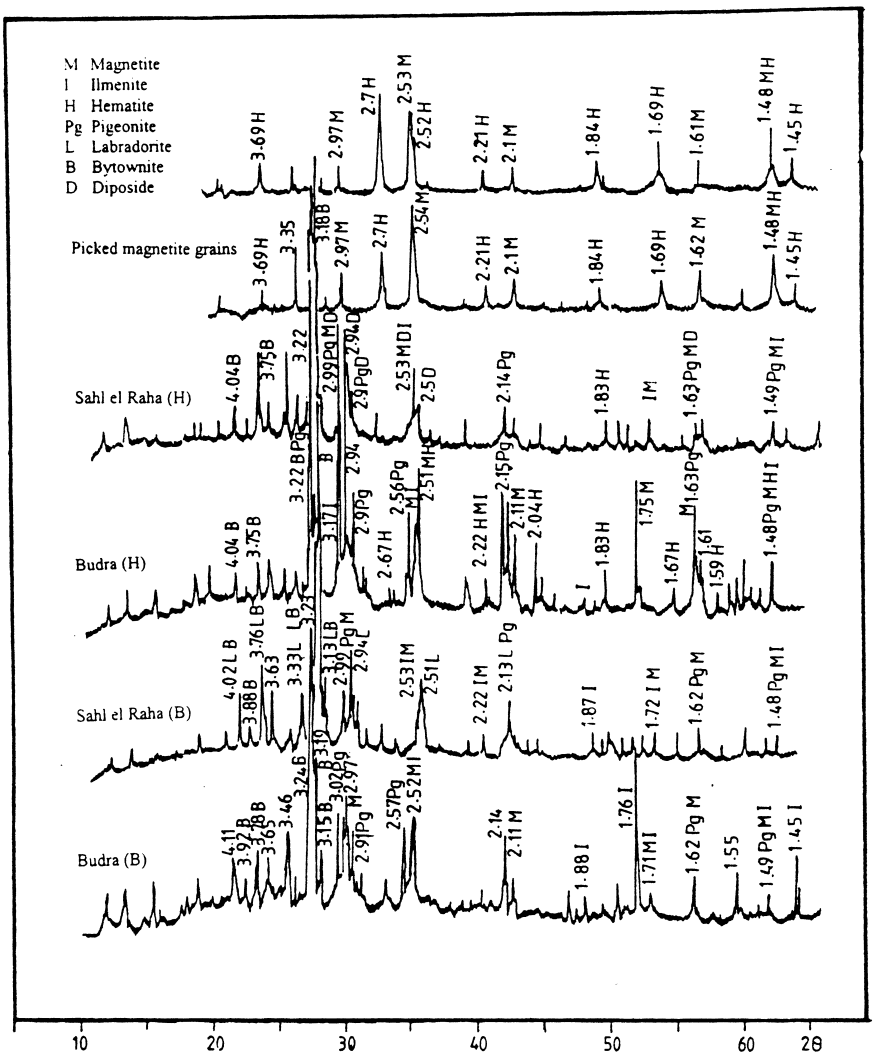


Fig. 2. X-ray diffraction patterns of some minerals from the studied basalt.

Shabiro and Branockis's (1962) methods of spectrophotometric (Si, Al, P), flame photometric (Na, K), titration (Fe^{2+}, Fe^{3+}, Mg, Ca) and gravimetric (LOI) techniques. The trace elements were analyzed by XRF technique using a Philips (PW 1410) X-ray spectrometer (Norrish and Chappell, 1966). All these chemical analyses were carried out in the Nuclear Materials Authority Laboratories, Cairo, Egypt and are listed in Table 1.

Zr, TiO_2, Y, P, V and Nb are conventionally regarded as relatively immobile during post-magmatic alteration owing to metamorphism, hydrothermal events and weathering (Pearce and Cann, 1973; Floyd and Winchester, 1975; Smith and Smith, 1976; Floyd and Ternary, 1979; Pearce, 1982). Immobility of these elements has some exceptions which could not be avoided (Finlow-Bates and Stumpfl, 1981; Schuessler et al., 1989). The information based on these immobile elements is greatly important in studying the volcanics under investigation.

Sahl el Raha rocks are enriched in Ti, V and Ba, and depleted in MgO compared with Budra basalt. The silica–total alkali plots (Saggerson and

Williams, 1964; Irvine and Baragar, 1971) of the studied samples indicate mildly alkaline affinity (Fig. 3A). Also, the Nb/Y versus Zr/TiO_2 diagram (Floyd and Winchester, 1978; Fig. 3B), nepheline- (Budra basalt) and olivine-normative (Raymond, 1995; Table 1) and $Y/Nb < 1.5$ (Pearce and Cann, 1973) of the investigated samples confirmed the alkaline affinity of those samples. The average *mg*-indices [$100 \times Mg/(Mg+Fe)$] are 38.37 and 31.70 for Budra and Sahl el Raha volcanics, respectively, indicating that these volcanics are fractionated (Oliveira and Tarney, 1995).

In order to define the tectonic setting of these volcanics, the Ti–Zr (Pearce, 1982; Fig. 3C) discrimination diagram was used. These plots suggest that the studied volcanics were erupted in a within-plate environment. Such within-plate basalts are enriched in most incompatible elements

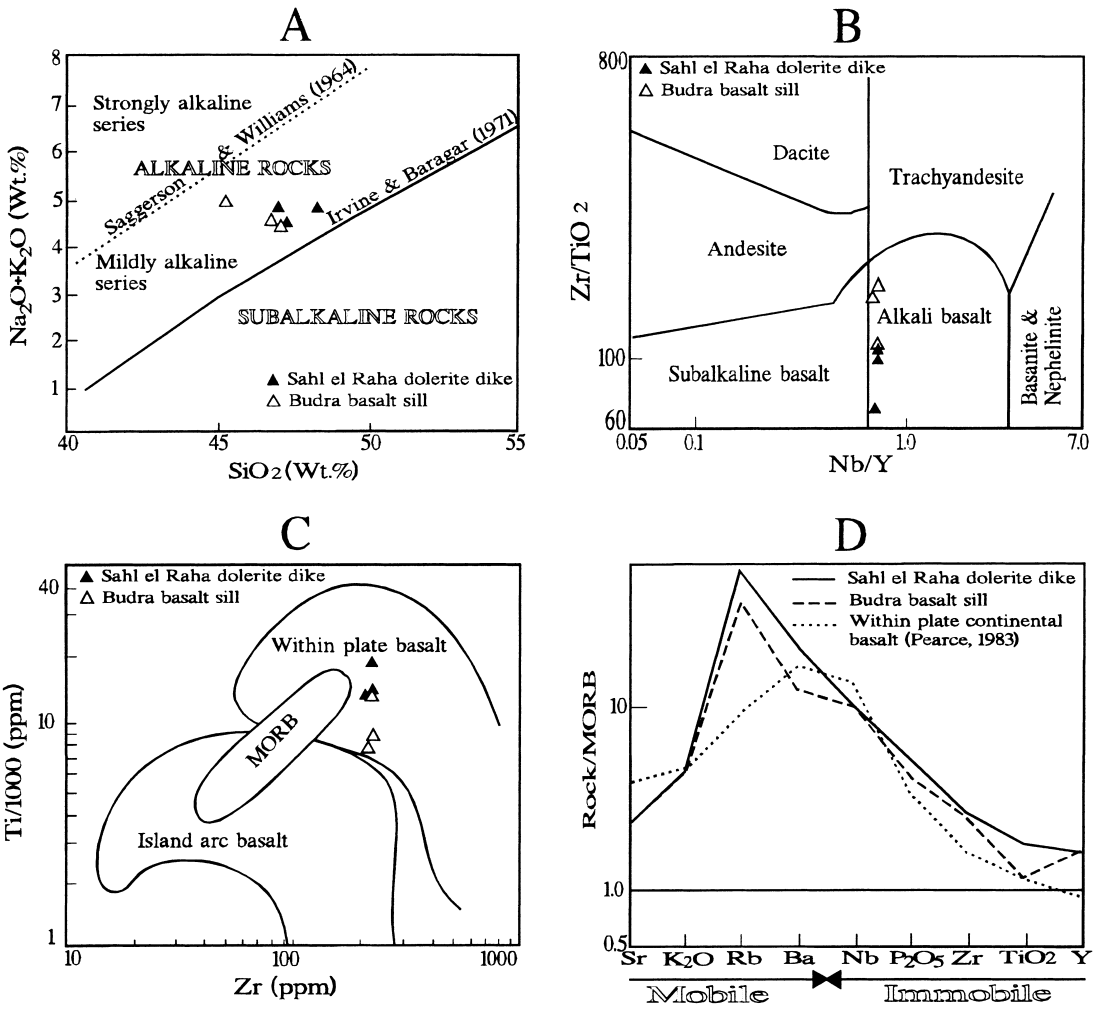


Fig. 3. Different discrimination diagrams of the studied samples. (A) Silica–total alkali plot. Solid line separates subalkaline from alkaline rock suites (Irvine and Baragar, 1971) and dashed line separates mildly alkaline from strongly alkaline series (Saggerson and Williams, 1964). (B) Nb/Y–Zr/TiO_2 diagram (Floyd and Winchester, 1978). (C) Zr–Ti/1000 tectonic setting discrimination diagram (Pearce, 1982). (D) MORB-normalized element abundance pattern of the studied rocks, normalization values are from Pearce (1983).

(Wilson, 1989) compared with MORB and show 'humped' MORB-normalized trace element patterns.

The MORB-normalized element abundance patterns of the studied volcanic rocks (Fig. 3D) match very well with the intra-continental plate basalts (Pearce, 1983) for most of the immobile and K elements, except Sr which is slightly depleted and Rb which showed more enrichment, comparable with the intra-continental plate basalts.

5. Paleomagnetic investigation

Six oriented block samples were collected from each volcanic occurrence at Wadi Budra and Sahl el Raha area. The locations of the sampling sites are shown in Fig. 1. The structural attitude (strike and dip) of Budra volcanic sill was carefully measured to allow for the structural rotation of magnetic vectors back to their pre-tilting horizontal position. After that cores of 2.5 cm in diameter were drilled in the laboratory from the blocks and cut into samples of 2.5 cm length, suitable for paleomagnetic measurements. The natural remanent magnetization (NRM) of all samples was measured using the Molspin magnetometer at the Department of Geological Sciences, University of Liverpool, UK. The initial intensities of the NRM were relatively high in both localities, ranging from 34 to 138 mA/m for the samples of Wadi Budra and 70 to 929 mA/m for those of Sahl el Raha area.

In this study, it was preferred to subject all the rock samples to a full range of alternating field (AF) demagnetization in steps up to 1000 Oe in order to avoid any anisotropic effects between the rock samples and to isolate the characteristic remanent magnetization from each rock sample. The AF demagnetization treatment was effective, yielded well-resolved, smooth and linear trajectories decaying to the origin and displayed all magnetic components carried by the rock samples.

Depending upon the analysis of the pilot samples using Zijderveld diagrams and stability data, the characteristic stable components of remanence (ChRMs), which persist until higher fields, were isolated. The directions of these characteristic components of magnetization were calculated using the principal component analysis program of Kirschvink (1980), based on the determination of the demagnetization trajectories heading towards the origin.

At the beginning of the progressive AF demagnetization, most of the treated samples showed remarkable erratic changes in magnetic direction and intensity. It is believed that this behavior was due to the presence of a soft component of remanent magnetization which is probably a secondary overprint of viscous remanence magnetization (VRM). This soft component was removed by about 100 Oe in most of the samples. At higher fields during AF demagnetization, the intensity of magnetization of most of the rock samples became too weak to be measured. Typical examples of the magnetic behavior during stepwise AF demagnetization of the samples are shown in Figs. 4 and 5.

6. Paleomagnetic results

6.1. Wadi Budra basalt

After applying the AF demagnetization and tilt correction (strike = 130° and dip = 25°SW) all sites showed a fair consistency, except site WB6, which was discarded because its respective direction of magnetization is far away from the mean. Giving unit weight to the remaining sites, the overall mean direction of magnetization was $D = 63.3°$, $I = 8.0°$ and $\alpha_{95} = 10.2°$. This yields a paleopole position at lat. 25°N, long. 133°E and $A_{95} = 8.6°$. The characteristic site mean directions of magnetization obtained after the treatment in the AF fields and the corresponding virtual pole positions (VGP), 95% circles of confidence (Fisher, 1953) and $N =$ number of analyzed samples are summarized in Table 2. The site mean directions and overall mean direction of magnetization are plotted in Fig. 6.

6.2. Sahl el Raha dolerite

After applying AF demagnetization, all samples from Sahl el Raha area showed normal magnetic polarity with fairly consistent directions. Giving unit weight to each site, the overall mean direction of ChRM magnetization of the sites was $D = 3°$, $I = 39°$ and $\alpha_{95} = 15.3°$ and the corresponding paleopole position was at lat. 83°E, long. 179°E

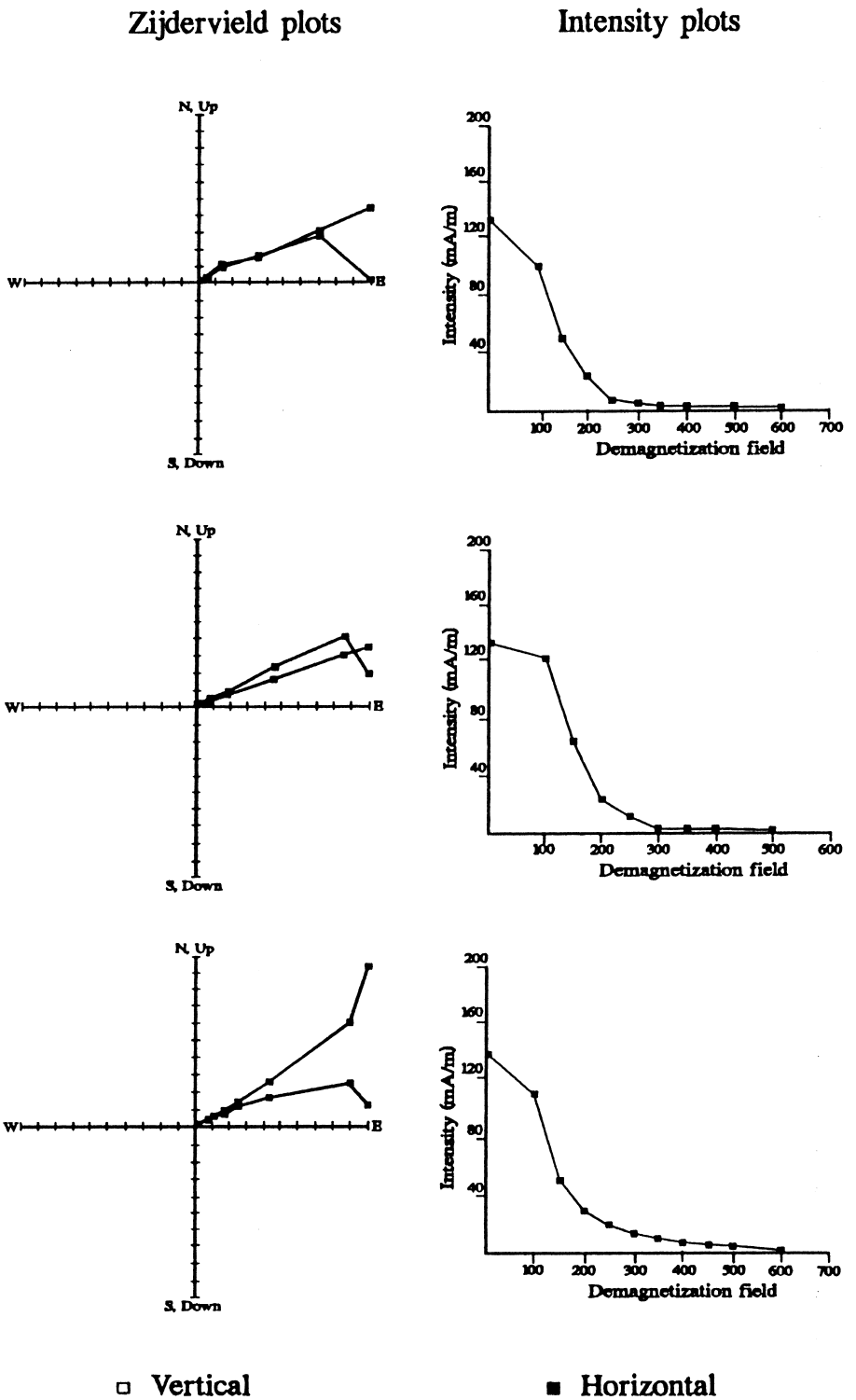


Fig. 4. Typical examples of the orthogonal vector diagrams (Zijderveld, 1967) and intensity decay plots of the Budra basalt samples.

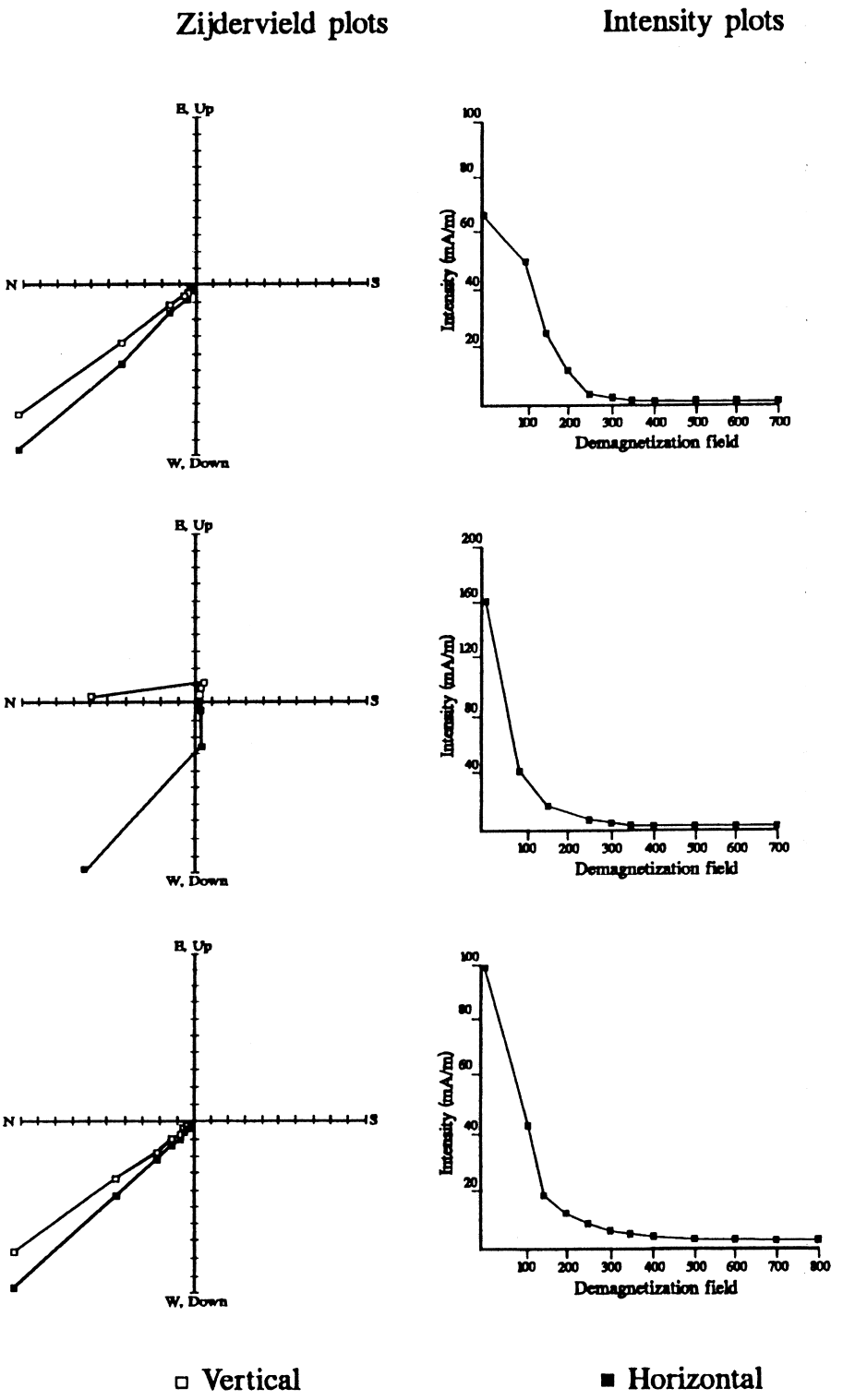


Fig. 5. Typical examples of the orthogonal vector diagrams (Zijderveld, 1967) and intensity decay plots of the Sahl el Raha dolerite samples.

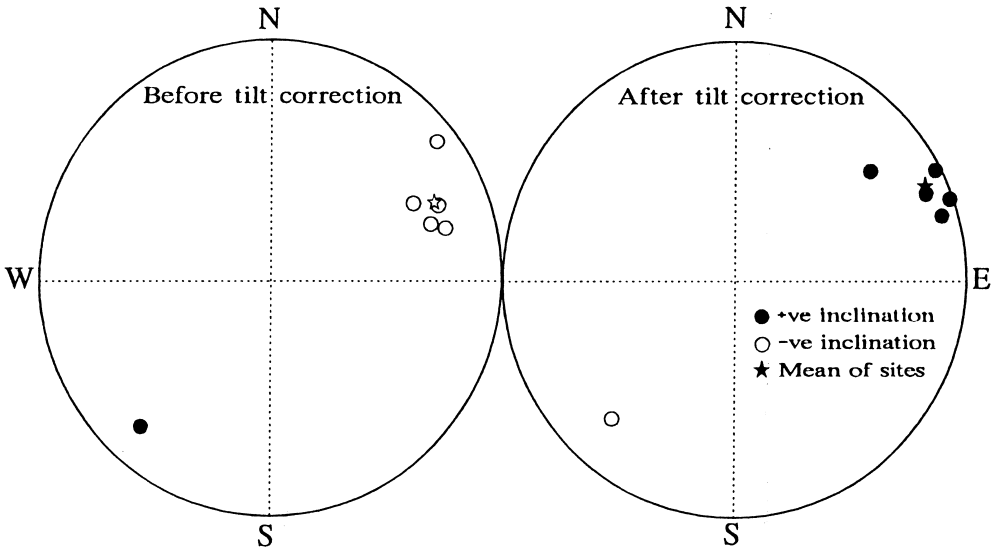


Fig. 6. Characteristic site mean directions of magnetization (before and after tilt correction) of the studied basalt of Wadi Budra.

Table 2
Cleaned paleomagnetic results of Wadi Budra basalt, southwest Sinai[a]

Site	Before tilt correction		After tilt correction		N	α_{95}	VGP		D_p	D_m
	Dec.	Inc.	Dec.	Inc.			Lat. N°	Long. E°		
WB 1	61.6	−23.0	60.3	2.0	5	7.3	26	137	3.7	7.3
WB 2	70.2	−20.6	68.5	3.2	5	5.5	19	132	2.8	5.5
WB 3	73.0	−16.5	71.9	6.7	4	10.0	17	129	5.1	10.1
WB 4	65.4	−16.2	64.8	8.3	6	8.6	24	132	4.4	8.7
WB 5	49.8	−6.8	50.0	19.1	5	3.0	39	135	1.6	3.1
WB 6	221.3	11.1	221.2	−14.8	4	12.1	Excluded		–	–
Mean	64.0	−17.0	63.0	8.0	5	10.2	25.0	133.0	$A_{95}=8.6°$	

[a] Dec., declination; Inc., inclination; N, number of samples; α_{95} and A_{95}, radius of the 95% circle of confidence (Fisher, 1953) for directions and poles, respectively; VGP, virtual geomagnetic pole; D_p and D_m, semi-axes of the 95% confidence oval.

Table 3
Cleaned paleomagnetic results of doleritic dike of Sahl el Raha, south Sinai[a]

Site	Dec.	Inc.	N	α_{95}	VGP		D_p	D_m
					Lat. N°	Long. E°		
SR 1	18.3	41.3	6	6.0	72	136	4.5	7.3
SR 2	20.0	39.4	6	10.1	70	138	7.2	12.0
SR 3	21.6	52.6	5	5.3	71	104	5.0	7.3
SR 4	354.0	40.6	7	7.0	82	260	7.2	10
SR 5	334.5	23.4	4	12.0	60	275	8.1	14
SR 6	1.2	28.8	6	5.5	77	208	6.1	8.2
Mean	3.0	39	6	15.3	83	179	$A_{95}=16.5°$	

[a] For key, see Table 2.

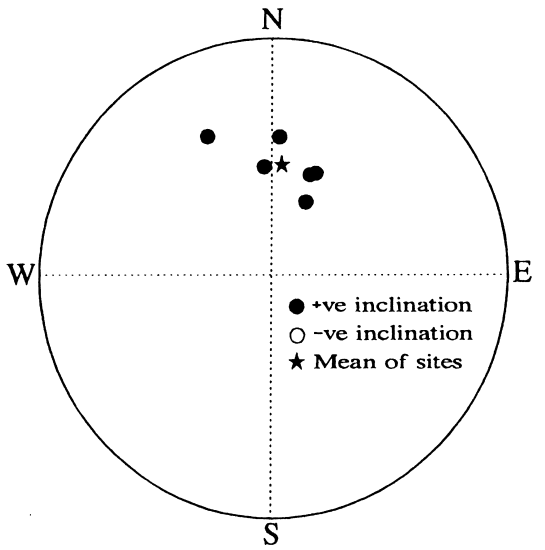


Fig. 7. Characteristic site mean directions of magnetization of the studied dolerite samples of Sahl el Raha.

and $A_{95}=16.5^{\circ}$. The site mean directions and overall mean direction of magnetization are listed in Table 3 and plotted in Fig. 7.

7. Discussion and conclusions

Two volcanic occurrences to some extent have similar petrographic characteristics and are composed essentially of andesine to bytownite plagioclase, pigeonite to diopside clinopyroxene and olivine. Magnetite is present as main opaque magnetic mineral (which is the main magnetization carrier in the studied samples), with a minor amount of ilmenite. The studied rocks are alkali olivine basalt at W. Budra and alkali olivine dolerite at Sahl el Raha. These rocks are related to a continental within-plate rift environment.

The rock samples are characterized by their high magnetic intensities (ranging from 34 to 929 mA/m). This high intensity more probably results from the abundance of unaltered magnetite as the main magnetic mineral and the opaque oxide minerals in the studied samples as revealed from the mineralogical investigations (Table 1 and Fig. 2). The magnetic remanence in the studied samples is also characterized by its high stability during AF demagnetization. The isolated ChRMs of the two volcanic occurrences are different from the present Earth's magnetic field and the axial–dipole field directions in studied localities (Dec.= 2.6° and Inc.=42°).

The paleomagnetic pole positions corresponding to the ChRMs of the studied rocks were compared with the reference paleomagnetic directions of Africa for similar ages (Table 4) given by Besse and Courtillot (1991) and Van der Voo (1993). From this comparison, it was found that the observed paleomagnetic direction of Budra basalt is not consistent with the expected paleomagnetic directions from stable Africa for a similar age (Fig. 8A). Also, there is a difference in inclination between that of Budra basalt and the mean paleomagnetic direction of Africa, which could suggest that Sinai was about 6° of latitude further south than in the African plate. Such a difference

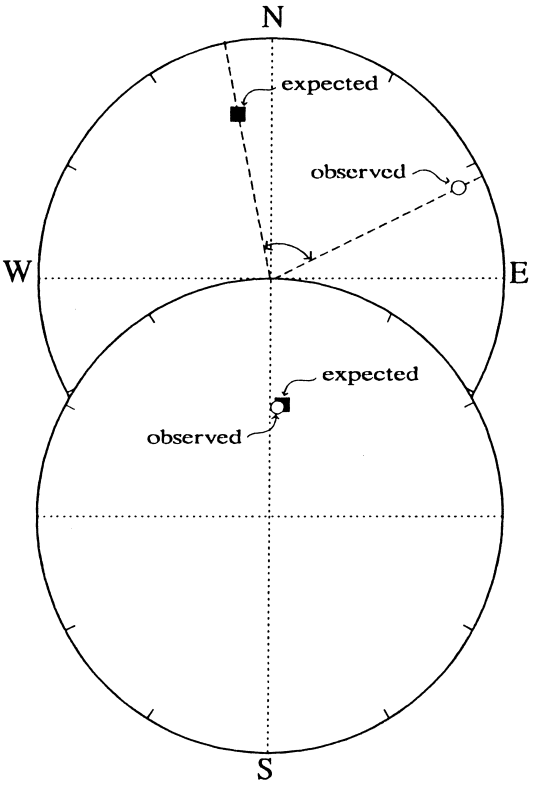


Fig. 8. Comparison of the site mean directions of magnetization and the expected field directions for Budra basalt (A) and Sahl el Raha dolerite (B).

Table 4
Present paleomagnetic results and reference directions (Besse and Courtillot, 1991)

Age	Direction			Pole position			Sampling locality	
	Dec.	Inc.	α_{95}	Lat. N°	Long. E°	A_{95}	Lat. N°	Long. E°
Wadi Budra	63.3	8.0	10.2	25.0	133.0	8.6	28°55′	33°20′
Reference	348.8	18.7		68.0	244.0	8.0		
Sahl el Raha	3.0	39.0	15.3	83.0	179.0	16.5	28°33′	33°55′
Reference	5.8	37.9		81.0	177.0	5.0		

is completely incompatible with the structural and lithologic matches on both sides of the Gulf of Suez.

Similar anomalous paleomagnetic directions have been reported from other areas along the Gulf of Suez. Wassif (1991) obtained anomalous results from the Mesozoic paleomagnetic results from two separate basaltic bodies at Gebel Farsh el Azraq and Wadi Budra in the eastern part of the Gulf of Suez, which were attributed to either local tectonic effects or inadequate averaging of secular variations. Moreover, Kafafy and Abdeldayem (1993) obtained other anomalous paleomagnetic directions from some Syrian Arcs from Sinai and the Eastern Desert around the Gulf of Suez, which were attributed to internal rigid deformation as a result of strike–slip faulting and block rotations as a result of the opening of the Gulf of Suez and isolation of Sinai subplate.

The tectonic rotations attributed to the effects of local tectonic movements have been widely reported throughout the southeastern Mediterranean region (Freund and Tarling, 1979; Ron et al., 1984; Ron, 1987; Heimann and Ron, 1993). So, the anomalous paleomagnetic direction from Wadi Budra basalt could be referred to local tectonic rotations affecting this area of study, especially the fact that the area is dissected by normal faults (Meshref, 1990) which could facilitate tectonic movement. In order to clarify this, the present paleomagnetic data were examined for local tectonic rotations by comparison with the mean African reference pole (Besse and Courtillot, 1991; listed in Table 4) and using the method outlined by Beck (1980) and modified by Demarest (1983). This comparison revealed that the difference between the observed pole positions from Budra volcanics and the reference pole of Africa was very large ($71 \pm 14^{\circ}$), which is not compatible with any tectonic rotations recorded for this part of Sinai. Such a large rotation angle is too great to be explained by an incomplete averaging out of the secular variation. This is because the Budra basaltic sill is a single cooling unit which consequently has a short duration of magmatic episode and rapid cooling. These uncertainties suggest that the calculated paleomagnetic pole from the Budra basaltic sill cannot genuinely be used for any tectonic conclusions.

On the other hand, the Tertiary volcanic activity phases were generally interpreted as being related to initial faulting in the Gulf of Suez, which was not active before Late Eocene times, about 40 Ma ago (Robson, 1971; Lowell and Genik, 1972; Berggren and Van Couvering, 1974). Thus, the early rift-forming events must have occurred between the Late Eocene and the beginning of the Miocene (Garfunkel and Bartov, 1977). In the present study, the paleomagnetic investigation is carried out on the Sant Catherine doleritic dike at Sahl el Raha area, southern Sinai as one occurrence of these Tertiary volcanics which is related to the opening of the Gulf of Suez and the Red Sea. Comparatively, the paleomagnetic results of Sahl el Raha dolerite are not significantly different from those Tertiary paleomagnetic results from Wadi Tayiba (Wassif, 1989), Wadi Matulla (Wassif, 1988) and Wadi Nukhul (Wassif, 1991). Also, the paleomagnetic direction from stable Africa (Table 4 and Fig. 8B) and the APWP of Africa (Fig. 9) given by Besse and Courtillot (1991). This comparison revealed that the paleomagnetic direction of Sahl el Raha is consistent with the part of the APWP of Africa for the time

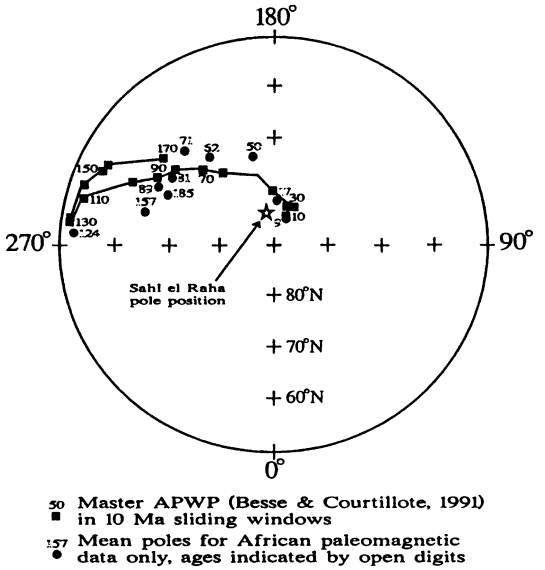


Fig. 9. Paleomagnetic pole position of Sahl el Raha compared with the African APWP (Besse and Courtillot, 1991).

period of 20 Ma, Oligo-Miocene age (Table 4). This pole represents new paleomagnetic data for the Tertiary period in Egypt, and is especially consistent with the K–Ar age of this volcanic intrusion (Steinitz et al., 1981). The large value of α_{95}, the radius of the 95% circle of confidence (Fisher, 1953), could be referred to the fact that the Sant Catherine doleritic dike at Sahl el Raha is a single cooling unit, and so could not completely average out the geomagnetic secular variations, and/or that the mean was calculated from only a few individual sites. Based on the relation between the magnetic inclination and paleolatitudes of the studied area, it is suggested that the Sant Catherine doleritic dike was erupted when Sinai was at latitude 21°, i.e. about 7° southward of its present position at the time of the eruption of this dike (20 Ma ago). This is explained by the regional northeastward movement of the African plate and the incorporated Arabia relative to Eurasia in response to the production of the southern oceans.

Finally, it could be concluded that the anomalous paleomagnetic results from the areas around the Gulf of Suez, either from Budra volcanics (present study) or from previous results (Kafafy and Abdeldayem, 1993; Wassif, 1991), indicate that these areas have been affected by local tectonic rotations as a consequence of the tectonic development of the Gulf of Suez, which is dominated by tensional stresses and block rotations along the normal faults which dissected the Gulf and the surrounding areas. In contrast, the compatibility of the paleomagnetic results of Sant Catherine doleritic dike with the similar paleomagnetic pole positions from Africa indicate that the dike was erupted after the rift-forming events occurred. It is obvious that the tectonic rotations in Sinai are local and restricted to the areas around the Gulf of Suez.

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